

Machine Learning for Congestion Management and Routability Prediction within FPGA Placement

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Placement for Field Programmable Gate Arrays (FPGAs) is one of the most important but time-consuming steps for achieving design closure. This article proposes the integration of three unique machine learning models into the state-of-the-art analytic placement tool GPlace3.0 with the aim of significantly reducing placement runtimes. The first model, MLCong, is based on linear regression and replaces the computationally expensive global router currently used in GPlace3.0 to estimate switch-level congestion. The second model, DLManage, is a convolutional encoder-decoder that uses heat maps based on the switch-level congestion estimates produced by MLCong to dynamically determine the amount of inflation to apply to each switch to resolve congestion. The third model, DLRoute, is a convolutional neural network that uses the previous heat maps to predict whether or not a placement solution is routable. Once a placement solution is determined to be routable, further optimization may be avoided, leading to improved runtimes. Experimental results obtained using 372 benchmarks provided by Xilinx Inc. show that when all three models are integrated into GPlace3.0, placement runtimes decrease by an average of 48%.

CCS Concepts: • **Hardware** → **Reconfigurable logic and FPGAs**; **Physical design (EDA)**; **Placement**; **Wire routing**; Partitioning and floorplanning;

Additional Key Words and Phrases: Placement, field programmable gate array, congestion, routing-aware, heterogeneous, UltraScale architecture

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1 INTRODUCTION

Many of the key optimization problems within the *Field Programmable Gate Array (FPGA)* design flow belong to the complexity classes NP-complete and NP-hard. Traditionally, *Computer-Aided Design (CAD)* developers have sought to solve these problems using solution methods, like heuristics, that forgo optimality in exchange for computational efficiency. However, as modern FPGAs push beyond an integration level of 30B transistors, not only are the size and complexity of FPGA

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